Microglia response to fatty acid binding protein-UCP2 axis in facilitating neuroinflammation

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Background

Introduction

Hypothalamic inflammation contributes to metabolic dysregulation and the onset of obesity [1, 2]. Dietary saturated fatty acids activate microglia via an NFκB mediated pathway to release pro-inflammatory cytokines, resulting in dysfunction or death of surrounding neurons [3, 4]. Fatty acid binding proteins (FABP)-4 is a lipid chaperone that regulates metabolic and inflammatory pathways in response to fatty acids [5]. Although inhibiting FABP4 in peripheral macrophages results in reduced obesity-induced inflammation via an uncoupling protein 2 (UCP2)-redox based mechanism [6], the expression of FABP4 and a potential FABP4-UCP2 axis in microglia is largely uncharacterized.

Hypothesis

We hypothesized that inhibition of FABP in microglia would attenuate lipopolysaccharide (LPS) inflammations through a UCP2 mediated mechanism (Figure 1).

Significance

- Chronic activation of microglia in the hypothalamus contributes to various metabolic diseases such as obesity.
- Diets rich in saturated fatty acids induce inflammation, which results in cell death and obesity.
- FABP4-UCP2 axis in microglia could decrease neuroinflammation, offering a novel anti-obesity therapeutic target.

Objectives

- Determine specific FABP isoform gene expression in microglial and brain tissue (Fig 1A-C).
- Determine changes in UCP2, arginase, and inducible nitric oxide synthase (iNOS) expression following inhibition of FABPs in the presence/absence of LPS in microglial (Fig 4A-C).

Methods

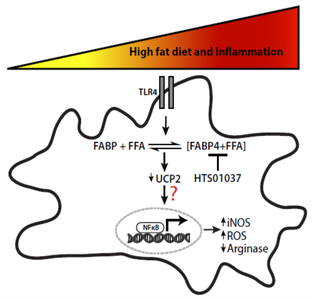
Cell culture and reagents: Immortalized murine microglial (BV2) were grown in Dulbecco's modified Eagle's medium (DMEM; Gibco) supplemented with 10% fetal bovine serum (Gibco) and 1% penicillin, streptomycin, and neomycin (Invitrogen) and maintained at 37°C with 5% CO2. Pan-FABP inhibitor (HTS01037;Caymen) was used at a final concentration of 30 μM. LPS (Sigma Aldrich) was used at a final concentration of 100 ng/ml in DMEM. [6]

BV2 microglial were seeded in 6 well plates and grown to ~80% confluency. Time points were based on Xu et al. [6]. For the first set of experiments, BV2 were treated with either vehicle control (DMSO), 15 μM or 30 μM HTS01037 for 24 h and collected with Trizol (Invitrogen) and stored at -20°C. For the second set of experiments, BV2 microglial were pretreated with either vehicle control or 30 μM HTS01037 for 3h and challenged with LPS or vehicle for 12 h. Cells were collected using Trizol and stored at -20°C until used.

Real time qRT-PCR: Total RNA was extracted from BV2 cells with the aid of Trizol and Qiagen RNeasy micro column as previously described [7, 8]. A final concentration of RNA (300 ng/ul) was determined using the 260/ 280 nm ratio (Nanodrop ND-8000). Gene was determined using qRT-PCR (Roche LightCycler 1.5) with SYBR Green detection. Primer sequences were generated using MacVector 15. Relative RNA levels were normalized to GAPDH using the Δ-ΔCT method [9]. Significant differences were determined using student's unpaired two-tailed t-test (GraphPad Prism 5). Amplification products were separated via electrophoresis on 3% agarose gels stained with SYBR green.

Results

Figure 1. Hypothesis schematic



Microglial express FABP4 and FABP5 but not FABP7

Figure 2. FABP expression in mouse whole brain tissue and BV2 microglial

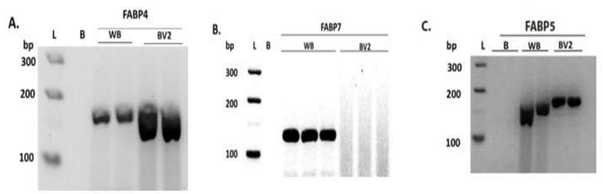


Figure 2: Fatty acid binding protein (FABP) is expressed in both whole brain (WB) and microglial. FABP7 (131 bp) is expressed in the WB (B), while FABP4 (174 bp) and FABP5 (168 bp) are expressed in the WB and microglia (A, C). L, ladder, B, water blank.

Inhibition of FABP in BV2 microglia increases UCP2 expression is dose dependent

Figure 3. UCP2 gene expression is increased following 24h treatment with HTS01037

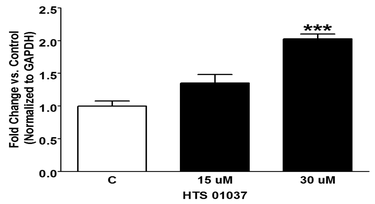


Figure 3: Fatty acid binding protein inhibitor HTS01037 increases UCP2 expression in BV2 microglia. Microglia were treated with 15 μM or 30 μM HTS 01037 or vehicle control (C) for 24 h and isolated for qRT-PCR. ***p<0.0001 vs. control.

Inhibition of FABP trends to reduce iNOS gene expression in microglial

Figure 4. Effects of FABP inhibitor HTS01037 on arginase and Inducible nitric oxide synthase (iNOS) expression in BV-2 microglia

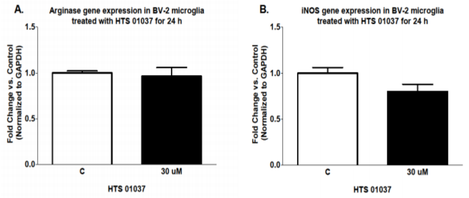


Figure 4: Effects of FABP inhibitor HTS01037 on arginase and iNOS expression in BV2 microglia. Microglia were treated with HTS 01037 or vehicle control (C) for 24h and isolated for qRT-PCR. HTS01037 does not change arginase expression (A) but trends to reduce iNOS expression (B; p=0.0826).

Inhibition of FABP stabilizes LPS induced pro-inflammatory gene expression

Figure 5. FABP inhibitor pretreatment prior to LPS challenge increases UCP2 and arginase expression but stabilizes iNOS expression

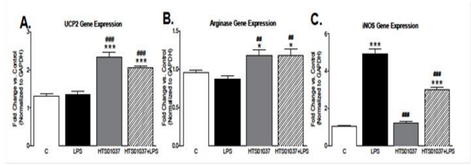


Figure 5: FABP inhibitor pretreatment prior to LPS challenge increases UCP2 and arginase expression but stabilizes iNOS expression. HTS01037 increases UCP2 (A) and arginase (B) gene expression compared to vehicle control and LPS only (*p<0.05 vs. control, *** p<0.0001 vs. control, ## p<0.001 vs. LPS, ### p<0.0001 vs. LPS). iNOS gene expression is stabilized with HTS01037 pretreatment (C; ***p<0.0001 vs. control, ### p<0.0001 vs. LPS).

Summary

Discussion

Exposure of the FABP inhibitor HTS01037 in the presence or absence of LPS as shown in figure 2-4 increased expression of UCP2 ($p<0.0001$ vs. control and LPS only) and arginase. This shows that FABP4 plays an important role in mediating inflammation by up regulating UCP2 expression and attenuating LPS induced pro-inflammatory response. Fatty acid binding proteins represent novel therapeutic targets within the central nervous system for anti-obesity treatment.

Future Directions

- Determine changes in reactive oxygen species following inhibition of FABPs in the presence/absence of LPS or the saturated fatty acid, palmitic acid (PA) in BV2 microglial.
- Determine changes in mitochondrial respiration following inhibition of FABPs in the presence/absence of LPS or PA in BV2 microglial.

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Acknowledgments

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